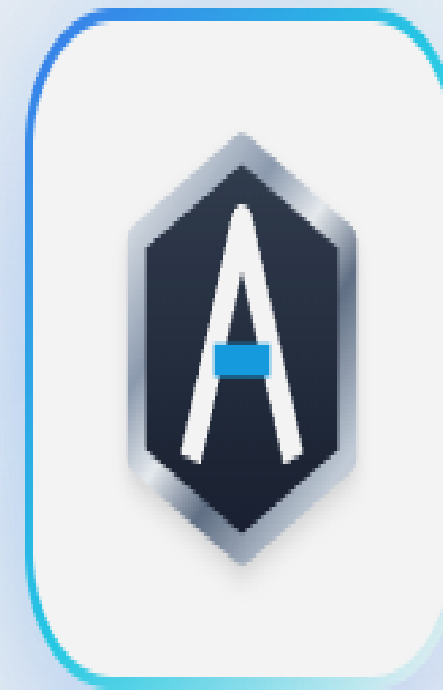




STRATEGIC TECHNICAL REPORT · 2026

Hard Engineering for the Sovereign Enterprise

SOC2 Secured

HIPAA / GDPR Ready

EU Compliance Ready

PhD Researcher · IITs · Advanced Data Systems & AI Strategy

adraca.io · contact@adraca.io

MANIFESTO

The Sovereign Vision

We replace experimental chaos with deterministic infrastructure. Adraca AI rejects technical debt in favor of industrial-grade reliability.

**PhD Analytics
Leadership:**

Founded on deep research from IITs — rigorous, not evangelical.

**\$52M+ Revenue
Impact:**

Proven track record at Accenture, PwC and global tier-1 firms.

Ironclad Sovereignty: Zero vendor lock-in for all critical enterprise workloads.

*"We do not build wrappers.
We engineer foundations."*

\$52M+

ENTERPRISE REVENUE IMPACT

4

PRODUCTION-GRADE AI PRODUCTS

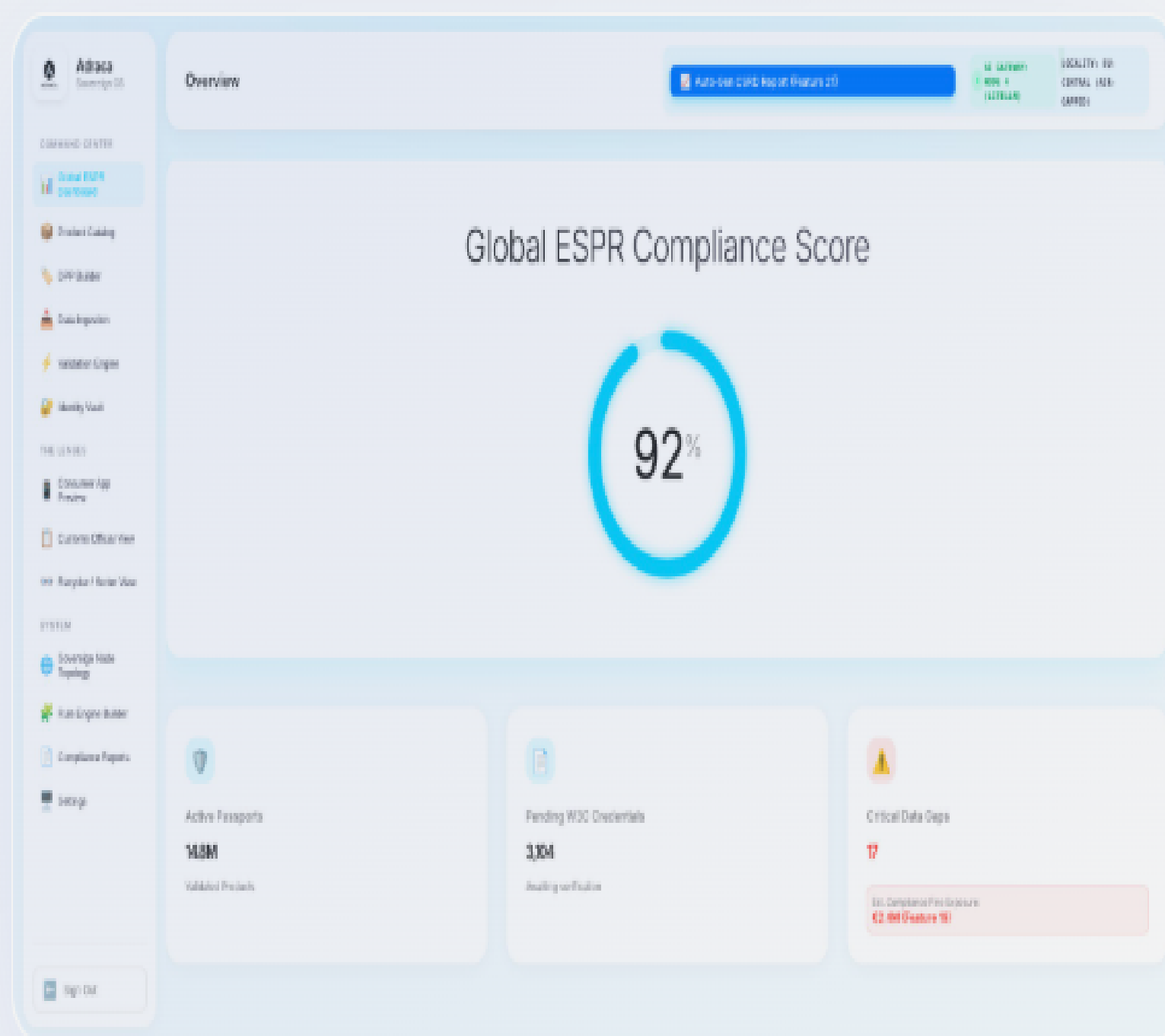
100%

ON-PREMISE · ZERO CLOUD LOCKOUT

Founded at the intersection of **research rigour** and **industrial engineering**, Adraca builds the infrastructure layer that enterprise AI requires to reach production.

Digital Product Passport (DPP)

The DPP Engine provides immutable lifecycle tracking for industrial assets, ensuring compliance with the EU's Ecodesign for Sustainable Products Regulation (ESPR).



THE SUSTAINABILITY PASSPORT

Think of this as a digital 'birth certificate' and 'passport' for every product. It tracks where materials came from and how to recycle them, helping you meet environmental laws automatically without the paperwork headache.

Real-time Circularity Metrics — across the full value chain

Multi-tier Supplier Traceability — with ZKP provenance

Automated Regulatory Reporting — born ESPR-compliant

```
dpp.verify('product_id')  
  .zkp_attest()  
  .emit_compliance_report();
```

TECHNICAL DEPTH

Zero-Knowledge Provenance

We utilise ZKP architectures to verify sustainability and compliance claims without exposing sensitive supplier pricing or trade secrets.

WHY THIS MATTERS

We prove your products are sustainable without revealing your secret recipes or supplier costs. It's 'Verification without Exposure' — keeping your competitive edge safe.



COMMAND CENTER

Global ESRP Dashboard

Product Catalog

DPP Builder

Data Ingestion

Validation Engine

Identity Vault

THE LENSES

Consumer App Preview

Customs Official View

Recycler / Sorter View

Dpp

Auto-Gen CSRD Report (Feature 21)

AI GATEWAY:
NODE 4
(LITELLM)

LOCALITY: EU-CENTRAL (AIR-GAPPED)

Back to Catalog

Digital Product Passport Builder

SKU: WRF6943

Product: Water-Resistant Field Jacket

STATUS: MINTED (ACTIVE)

ESPR Data Schema

100% Verified

Material Composition

Primary: 100% Organic Cotton
Coating: PFC-free DWR poly

Supply Chain Origin

Sourced: IT (Italy) - Farm ID 92-A
Manufactured: PT (Portugal) - Facility 03

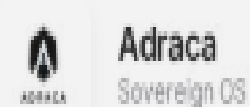


Recycling Intelligence

Closing the loop with automated material identification. The Recycler Engine maps individual components to recovery protocols in real time.

WHY THIS MATTERS

Automatically identifies materials in old products so they can be reused or recycled. This turns waste into value and proves your company's commitment to the circular economy.



COMMAND CENTER

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Recycler

Auto-Gen CSRD Report (Feature 21)

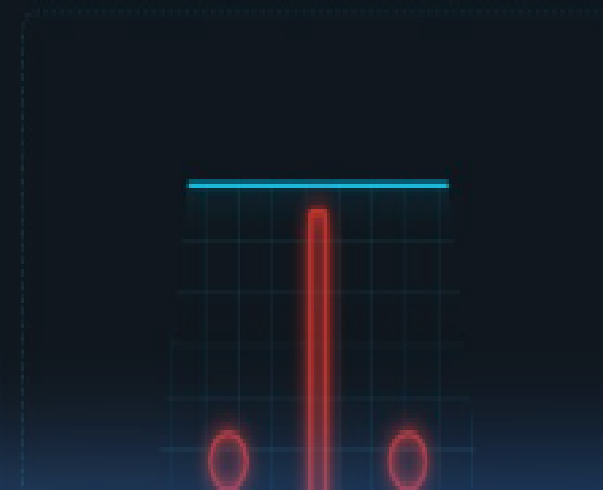
AI GATEWAY:
NODE 4
(LITELLM)

LOCALITY: EU-
CENTRAL (AIR-
GAPPED)

AR | TEXTILE ASSISTANT

Temp: 19°C Humidity: 42% Battery: 88%

ITEM: HERITAGE FIELD JACKET [ID: WRF0043]



MATERIAL COMPOSITION

95% Cotton

5% Elastane

Lens: Recycler/Sorter (AR)

This view imagines the JSON-LD data payload projected into an industrial AR headset on the factory or sorting floor.

- High Legibility: Neon cyan against dark/transparent backgrounds for low-light industrial environments.



ENTERPRISE AI

Sovereign Search Engine

Our Hybrid Fusion Core merges traditional keyword relevance with advanced vector-based semantic discovery — running 100% within your perimeter.

INTELLIGENT KNOWLEDGE DISCOVERY

A search engine that understands your company's documents like a human expert. It finds exactly what you need even without 'perfect' keywords—and it stays 100% on your servers so your data never leaves the building.

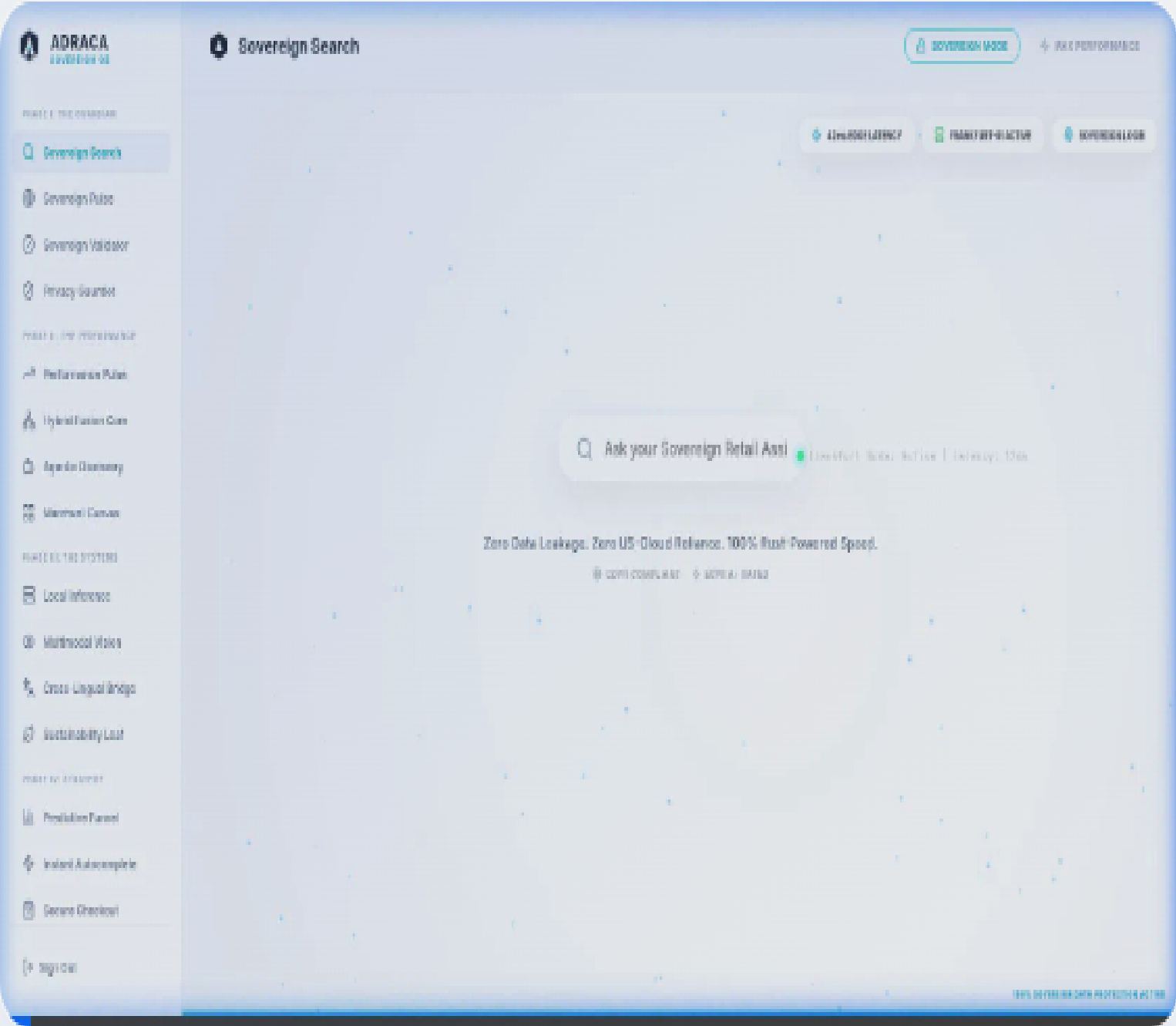
100% Local Inference — Your data never leaves your perimeter.

Cross-Lingual Bridge — Seamless discovery across global documentation.

Privacy Gauntlet — Automatic PII redaction and security filtering.

10ms

P99 SEMANTIC QUERY LATENCY



COGNITIVE UTILITY

Cross-Lingual Knowledge Discovery

Query in English, discover in Japanese. Our zero-data-leakage translation layer ensures global documentation is searchable without external API calls.

WHY THIS MATTERS

Breaks language barriers across your global offices. Search in English to find results originally written in Japanese or German—instantly, without using risky external translation tools.



PHASE I: THE GUARDIAN

🔍 Sovereign Search

🌐 Sovereign Pulse

✅ Sovereign Validator

🛡️ Privacy Gauntlet

PHASE II: THE PERFORMANCE

📈 Performance Pulse

🧩 Hybrid Fusion Core

📦 Agentic Discovery

🗺️ Merchant Canvas

PHASE III: THE SYSTEMS

Cross-Lingual Bridge

🔒 SOVEREIGN MODE

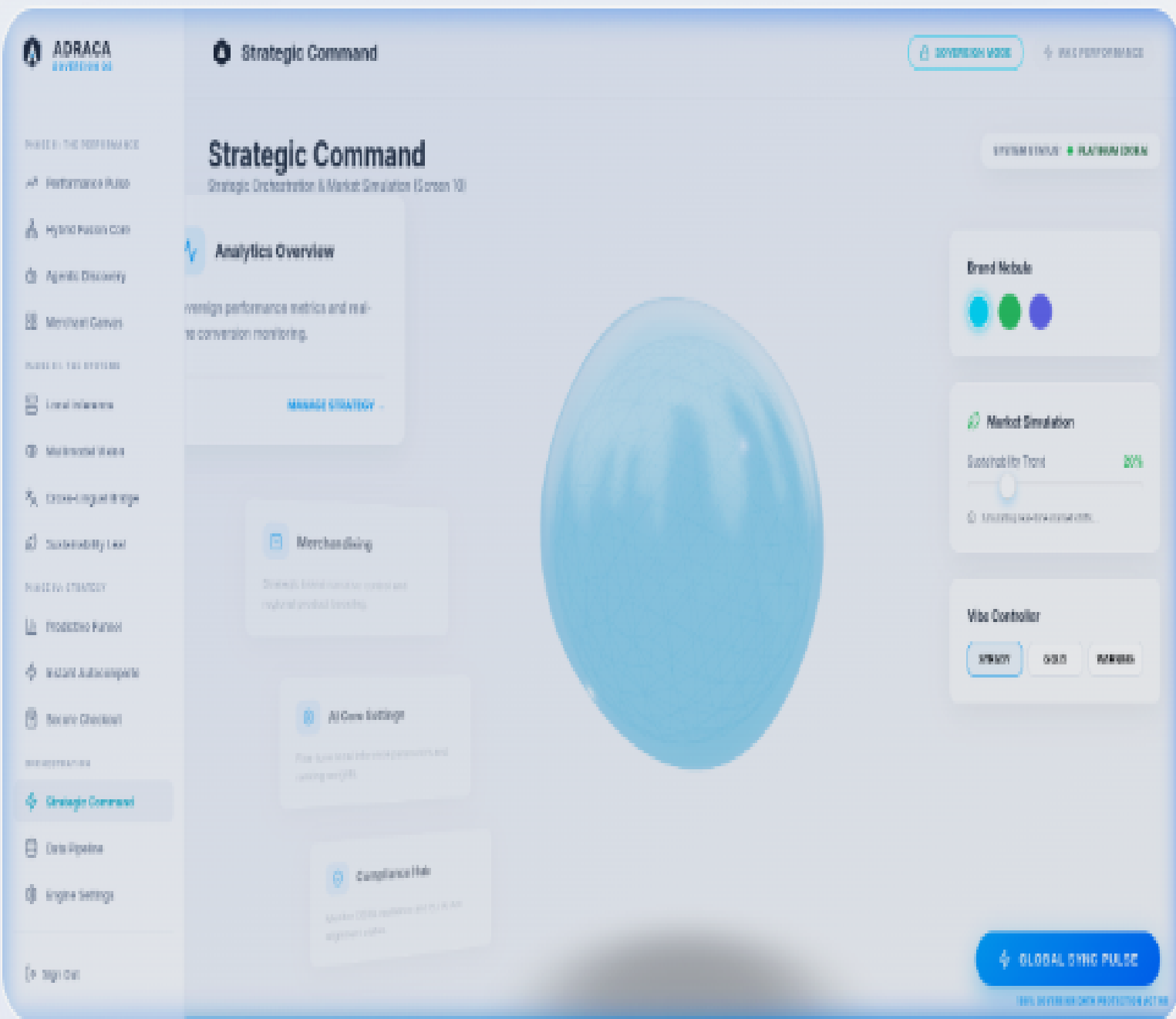
⚡ MAX PERFORMANCE

Cross-Lingual Bridge

EU Digital Single Market Semantic Mapping

🌐 5 LANGUAGES ACTIVE

⚡ 18ms LATENCY



ADMINISTRATION

Strategic Command Center

Complete control over your internal knowledge base. Administrators tune relevance, audit agentic trails, and monitor data health in real time.

BUSINESS IMPACT

Gives your team a simple control panel to manage how AI finds information. You can easily adjust what the AI prioritizes and see exactly who is accessing what data.

Dynamic Relevance Tuning

Agentic Discovery Audit Logs

Per-User Access Granularity

ACCESS MODEL

RBAC

Zero Trust

E2E Encrypted

Synthetic Patient AI

Accelerating clinical validation while preserving 100% data privacy. Our engine generates deterministic personas for medical system stress testing.

SAFE MEDICAL TESTING

We create 'digital twins' of patients that act like real people but contain zero private info. This lets medical companies test systems faster and cheaper with zero risk of leaking real patient data.

HIPAA & GDPR Compliant — by architecture, not policy

Deterministic Persona Generation — for reproducible trials

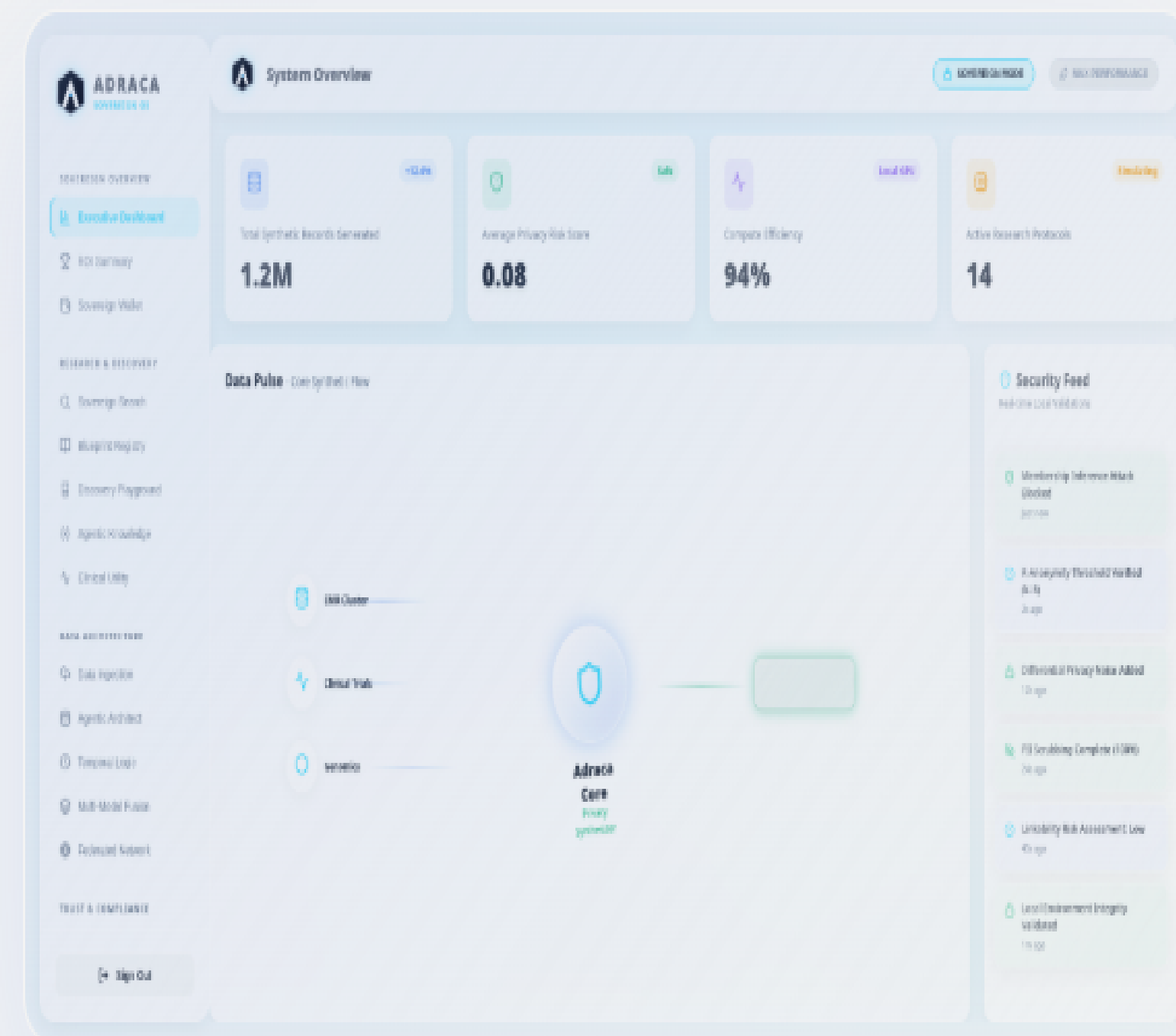
System Edge-Case Validation — at scale without real data

00

SYNTHETIC PATIENTS

0%

REAL PII EXPOSED



PRIVACY CORE

Privacy Perimeter Enforcement

Our Privacy Gauntlet ensures that not a single byte of real PII enters the synthetic generation loop. We utilise differential privacy at the edge.

WHY THIS MATTERS

An invisible shield that makes it mathematically impossible for real patient identities to be exposed. We use 'Differential Privacy' to ensure your data stays anonymous even during complex testing.



Q Sovereign Search

📖 Blueprint Registry

📄 Discovery Playground

🔗 Agentic Knowledge

🚀 Clinical Utility

DATA ARCHITECTURE

🔄 Data Ingestion

🏗️ Agentic Architect

🕒 Temporal Logic



Privacy Gauntlet Validation

🔒 SOVEREIGN MODE

⚙️ MAX PERFORMANCE

▶ Run Privacy Gauntlet

Attack Vectors

Automated DCR Verification

Membership Inference

Detects if real patient data was memorized.

Attribute Linkage

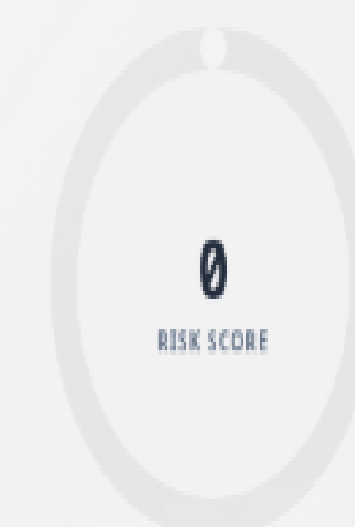
Attempts to guess hidden quasi-identifiers.

Singling Out

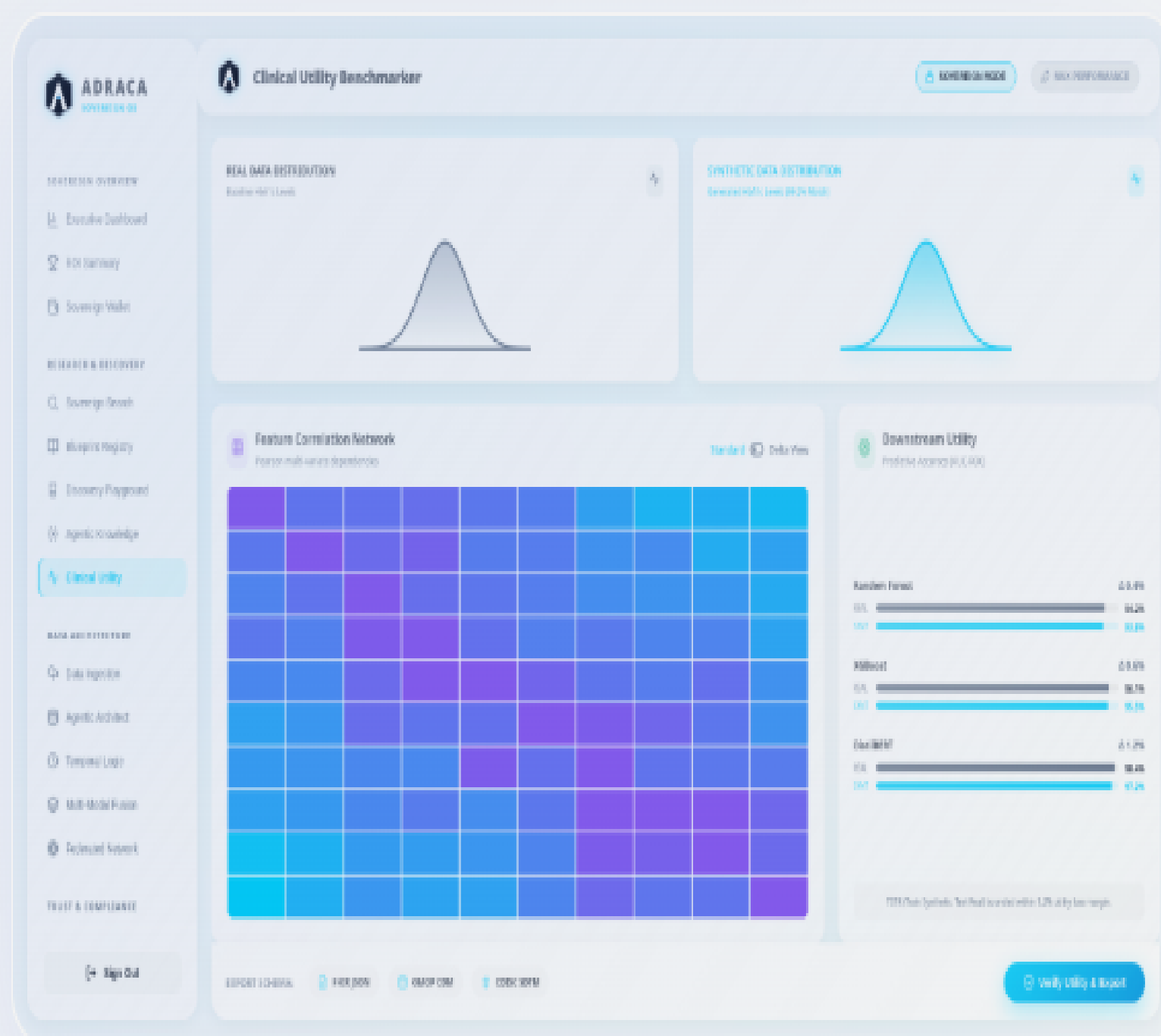
Identifies statistically unique synthetic records.

Re-identification Risk

DCR & MIA Valuation



COMPLIANCE STATUS



ANALYTICS

Clinical Utility & ROI

Drastically reduce patient recruitment costs and data preparation cycles with real-time visibility into the value delivered across the trial lifecycle.

BUSINESS IMPACT

Speeds up the time it takes to get new medical treatments ready for testing. By using AI patients first, you can find and fix problems early, saving millions in trial costs.

40%

AVG. GAIN IN TRIAL READINESS EFFICIENCY

60%

REDUCTION IN DATA PREPARATION TIME

Benchmarked across 3 Phase II clinical trials. Results validated with independent biostatistics review.

Sovereign Validator

Engineered for the DORA era. Automate the verification of service availability and contract SLA compliance across your entire vendor ecosystem.

AUTOMATED DIGITAL GUARD

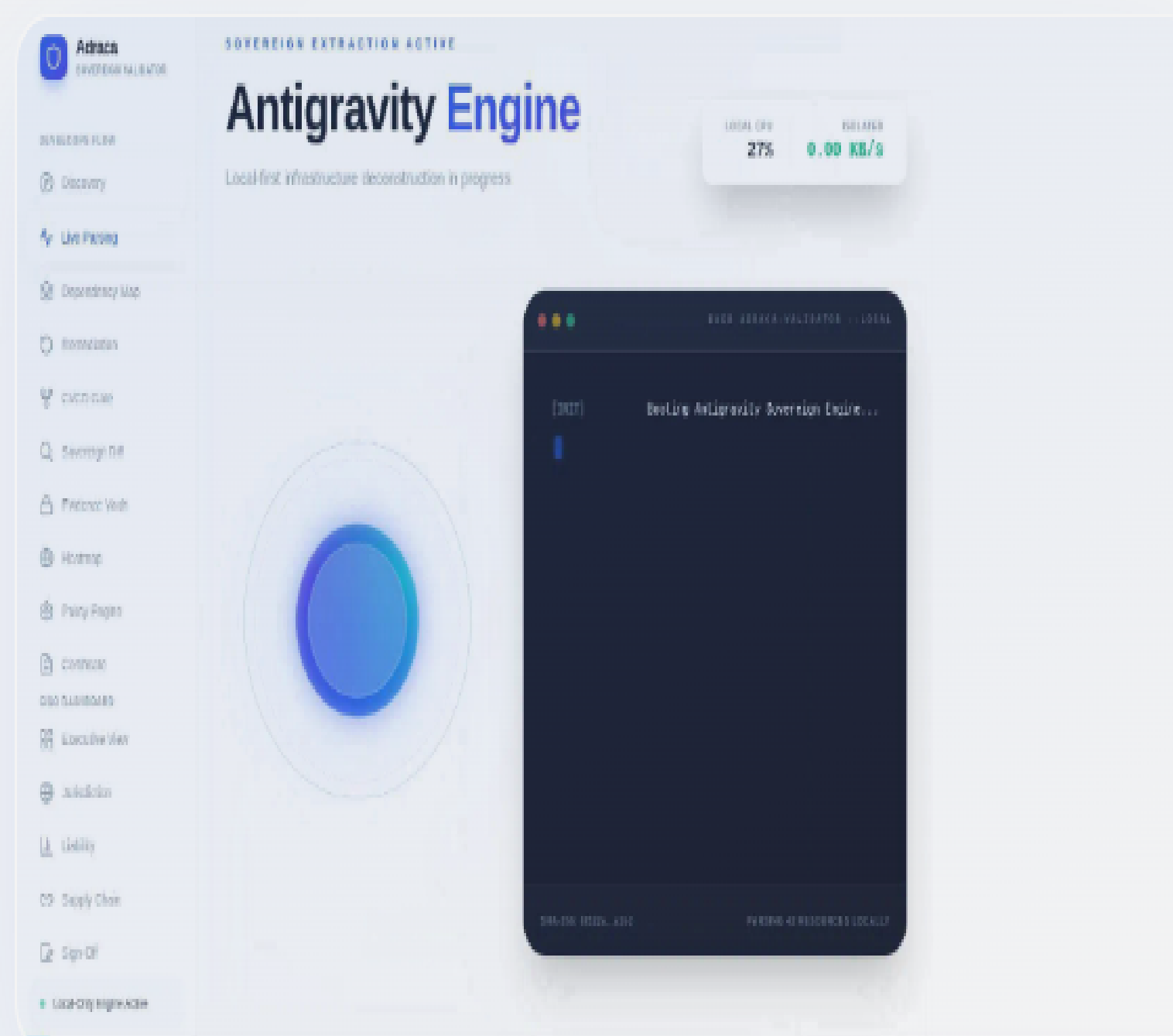
A 24/7 watchdog for your business. It monitors tech suppliers (like Cloud providers) to ensure they deliver what they promised. It alerts you the moment a service fails, helping you avoid outages and stay compliant.

Automated Vendor Status Scraping — with real-time alerting

PDF SLA Contract Parsing — using LLM-powered NLP

Immutable Compliance Audit Trails — ready for regulator review

```
validator.scan('aws.com')  
  .parse_sla(contract.pdf)  
  .emit_audit_trail();
```



DORA READINESS

Strategic Risk Landscapes

Visualise vendor risk across your entire portfolio. Automatically detect SLA breaches before they impact your core operations.

WHY THIS MATTERS

A 'warning map' for your technical risks. See at a glance which parts of your business are vulnerable to supplier failures and get ahead of problems before they impact your customers.



DEVSECOPS FLOW

Discovery

Live Parsing

Dependency Map

Remediation

CI/CD Gate

Sovereign Diff

Evidence Vault

Heatmap

CONCENTRATION RISK

Sovereign Heatmap

Visualizing regional density and single-point-of-failure risks (DORA Art. 12)

EUROPEAN OVERLAY ACTIVE

Resilience Score

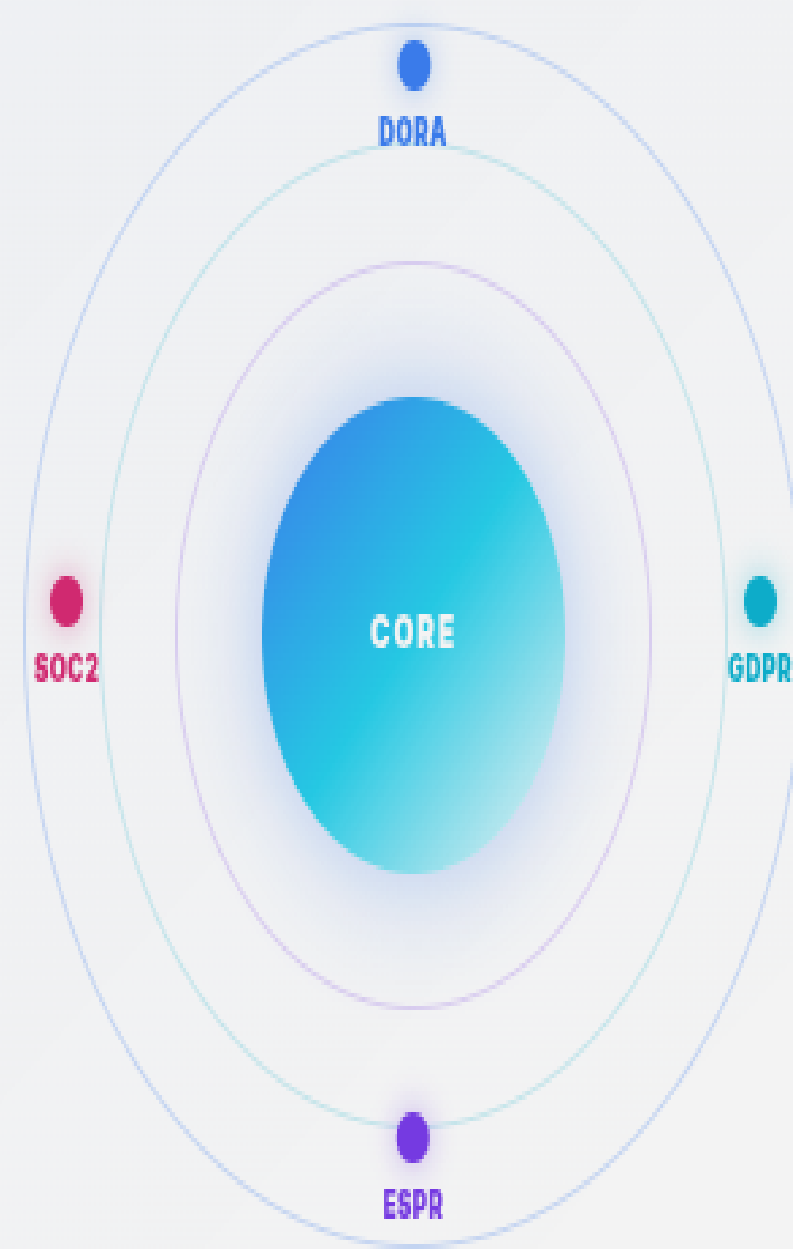
DORA COMPLIANCE AUDIT

DIVERSITY INDEX

68%

CONCENTRATION WARNING

CRITICAL: 82% density in Frankfurt creates a single-point-of-failure risk under



CORE INFRASTRUCTURE

Sovereign Landing Zones

We treat infrastructure as software. Terraform and Policy-as-Code vend cloud environments that are born compliant — not patched compliant.

Regional Data Residency — enforcement at the platform layer

Zero-Configuration Drift — via declarative state management

Audit-Ready EU Transparency — with real-time compliance graphs

```
terraform plan -d DORA_Landing_Zone_v3  
✓ 0 compliance violations detected
```

DATA STRATEGY

Declarative Pipelines

The Medallion Discipline: raw data is a liability; engineered data is an asset. We architect Bronze-Silver-Gold structures on Snowflake and Databricks.

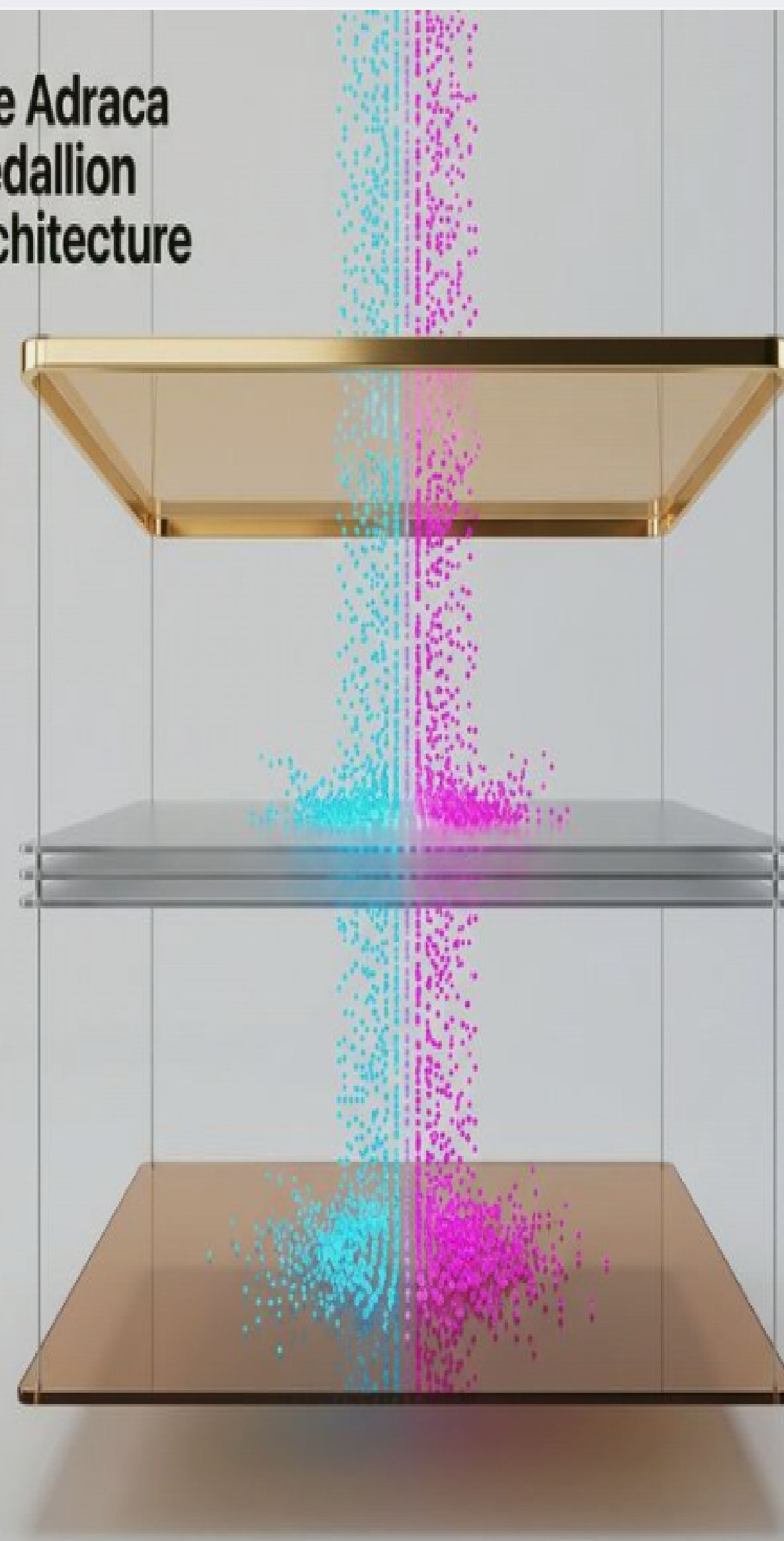
Automated Lineage Graphs — always know where data came from

Strict Modular dbt Logic — no spaghetti transformations

CI/CD-Enabled Reproduction — data as code, not as chaos

The Adraca Medallion Architecture

VisionOS



SOVEREIGN ENGINEERING

Product Leadership

Built by engineers who have operated at industrial scale — not theorists chasing demos.



Dipti A. Khaparde
Founder & CEO

100% Vision Execution

SPECIALTY

Scientific Computing & Industrial Processes

Expert in Chemical Engineering and complex process simulation. Dipti leads the translation of physical industrial constraints into digital twin architectures.



Vaishali Aathavale
Founder

12+ Years of Experience

SPECIALTY

High-Performance Computing

Leading HPC strategies and patent-pending innovations in distributed computing architectures for sovereign enterprise workloads.



Abhishek K.
CTO

Ph.D. Candidate (IITs)

SPECIALTY

Sovereign Data Infrastructures

Senior Architect aligning technical stacks with EU Data Act & DORA mandates. Architects the 'Third Way' of digital sovereignty.

NEXT STEPS

Initialize Partnership

We begin with a **Free Sovereign Audit** — a technical review of your infrastructure against DORA, ESPR, and data sovereignty standards.

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LOCATION

Europe / India Operations

SOC2 Secured

HIPAA / GDPR Ready

Estonian e-Residency